MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 12ATP28

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus, three beams, with aluminized screen.

Outline - Figure 1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec1	Eb1	Ec2	Eb2	Eb 3
Unit:	V	Vdc	Vdc	Vdc	Vdc	Vdc
Maximum:	6.3+10%	0	2000	6000	6000	15,000
Minimum:	6.3-10%	-200	----	----	----	------
TEST CONDITIONS: Note 1	6.3	Adj.	Focus	5000	Adj.	10,000

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	ed	Rg	Zd	Ehk	Eb3/Eb2
Unit:	v	Meg	Meg	Vdc	Ratio
Maximum:	1250	1.5	1.0	+180	2.0
Minimum:	----	---	Note 2	-180	Note 3
TEST CONDITIONS: Note 1	----	---	1.0	----	2.0

GENERAL:

Qualification - Required
Container drop - Method 1136

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification					
5101	Alinement, neck and bulb	---	---	---	4.375	in.
5216	Cathode illumination	---	---	---	---	---
5248	Deflection factor uniformity	1D2	---	---	2	σ_x
		3D4	---	---	2-1/2	σ_c
5250	Interaction factor		---	---	$6x10^{-5}$	in/Vdc
1026	Low temperature operation	---	---	-65	---	°C
1026	High temperature operation		---	---	+85	°C
1331	Direct interelectrode capacitances	Cathode to all	C	---	7.0	pF
		Grid 1 to all	C	---	7.0	pF
		D1 to D2	C	---	5.0	pF
		D3 to D4	C	---	5.0	pF
		D1 to all	C	---	9.9	pF
		D2 to all	C	---	9.9	pF
		D3 to all	C	---	8.6	pF
		D4 to all	C	---	8.6	pF
1141	Pressure	---	---	---	---	---
5111	Vibration	---	Width	---	2	mm
	Quality conformance inspection, part 1 See note 13					
5201	Cathode current	Ib3=5uAdc	IK	---	500	μAdc
5201	Voltage breakdown	---	---	---	---	---
5206	Gas	Ib3=5uAdc Note 12	---	---	---	---
5101	Alinement base	+1D2, Base key	---	---	±10	degrees
5101	Alinement side terminal	+1D2	---	---	±5	degrees
5101	Alinement side terminal and base	Base key	---	---	---	---
5101	Angle between traces	Note 4	---	---	1	degrees
		Guns A&B	---	89	91	degrees
		Gun C	---	88	92	degrees
5010	Trace alinement	Note 5	---	---	---	---
5216	Spray emission	Ec2=6000 Vdc; Eb2=6000 Vdc; Eb3=15,000 Vdc	---	---	---	---
5106	Blemishes	---	---	---	---	---
5223	Modulation	Ib3=5uAdc	Δ Ec1	15	30	Vdc

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 cont' (See note 13)					
5226	Line width A	Ib3=5uAdc; Note 6	---	---	0.013	in.
5226	Line width B	Ib3=5uAdc Notes 6 and 7	---	---	0.018	in.
5231	Spot position	Note 8	---	---	7/16	in.
5251	Spot displacement	---	---	---	0.2	in.
5236	Grid cutoff voltage	---	Ec1	-64	-86	Vdc
5246	Focusing voltage	---	Eb1	---	1725	Vdc
5246	Focusing voltage	---	Eb1	1275	---	Vdc
5248	Deflection factor	1D2, Guns A and B, Note 9	DF	103	127	Vdc/in
5248	Deflection factor	3D4, Guns A and B	DF	81	99	Vdc/in
5248	Deflection factor	1D2, Gun C Note 9	DF	103	127	Vdc/in
5248	Deflection factor	3D4, Gun C	DF	67	83	Vdc/in
5251	Grid #1 leakage	---	Ic1	---	3	μAdc
5251	Accelerator	---	Ic2	---	5	μAdc
5251	Anode #2 leakage current	---	Ib2	---	5	μAdc
---	Useful scan	Note 10	---	---	---	---
5103	Pattern distortion	Note 11	---	---	---	---
	Quality conformance inspection, part 2					
1301	Heater current	---	If	1620	1980	mA
5201	Electrode current (anode)	Ec1=0	Ib1	-15	+10	μAdc
5101	Alinement base and neck	---	---	---	---	---
5221	Right output	Ib3=5uAdc	---	25	---	ftL
5251	Heater-cathode leakage	---	---	---	10	μAdc
1101	Base, cap, and insert secureness	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
---	Life test	Group C; Ec2=Eb2=600 Vdc; Eb3=15,000 Vdc; Ib3=5uAdc	---	---	---	---
---	Life-test end point (500 hrs)	Ib3=5uAdc				
		Modulation	ΔEC1	---	30	Vdc
		Line width A	---	---	0.013	in.
		Line width B	---	---	0.018	in.

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NOTES:

1. All tests except vibration and interaction factor shall be made on each unit separately.

2. It is recommended that the deflection electrode circuit resistances be 1 megohm or less and approximately equal; otherwise beam shift at high drives can be expected. Higher resistance values up to 5 megohms may be used for low beam current operation.

3. This tube is designed for optimum performance when operating at an Eb3/Eb2 ratio of 2.0. Operation at other ratios of Eb3/Eb2 may result in changes in deflection uniformity and pattern distortion.

4. With the horizontal traces of guns A and B passing through the tube face center, and the horizontal trace of gun C passing through its undeflected spot point, all three traces shall be within 1 degree maximum angle.

5. Corresponding horizontal traces of each unit shall be within 1 degree of each other.

6. With focus (Eb1) and astigmatic (Eb2) voltages adjusted for the best center line width, line widths "A" and "B" shall be not greater than 0.013 and 0.018 inch, respectively. The specified spot position for each gun shall be used to locate the points for line widths "A" and "B".

   At no point within the specified scan area of any gun shall the line width exceed 0.080 inch with focus and astigmatic voltages adjusted for the best center line width.

   At no point within the specified scan area of any gun shall the line width exceed 0.030 inch when compromise focus and astigmatism correction are used.

7. Line width "B" shall be measured at 1.6 inches from the undeflected spot position for guns A and B and at 0.8 inch for gun C. The test shall then be repeated for guns A and B with the roles of horizontal and vertical plates reversed so that measurements shall be made at 5 points for each gun; center, 1.6 inches to right and left of center, and 1.6 inches above and below center.

8. Each spot shall fall within a circle of 7/16 inch radius, centered about the specified spot position (see figure 2).

9. The 1D2 deflection factor of each gun shall be within 5 percent of that of the other guns.

10. The useful scan of each gun shall be as shown on figure 2.

11. The trace from each gun, when deflected with a rectangular raster so that its widest points just touch the outside rectangle as shown on figure 3, shall fall entirely within the outide rectangle, but entirely outside the inscribed rectangle. Note that the horizontal trace from gun A, when passing through the tube face center, and when deflected to the full useful width, falls entirely within a rectangle 6.95 inches wide and 1/16 inch high.

12. Test shall be performed at the conclusion of the holding period.

13. Inspection shall be 100 percent.

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

(Porject 5960-N830)

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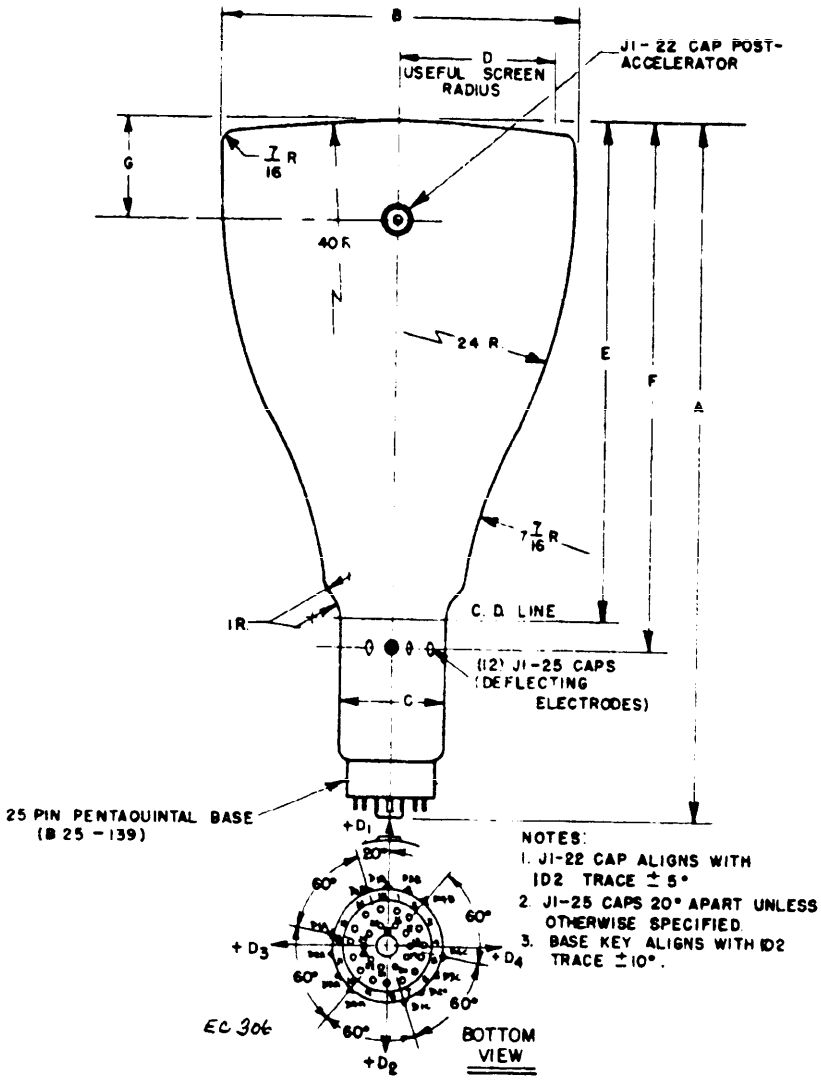
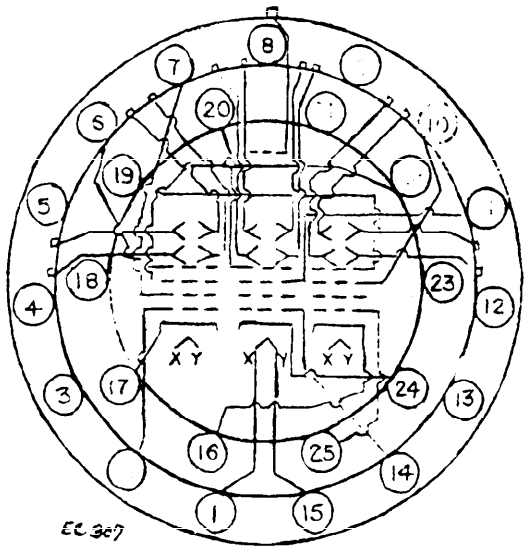


Figure 1 - Outline dimensions of tube type 12ATP28.

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ELEMENT	PIN NUMBERS UNIT A	UNIT B	UNIT C
HEATER (COMMON)	1	1	1
HEATER (COMMON)	15	15	15
CATHODE	17	24	16
FOCUSING ELECTRODE	22	11	10
ACCELERATOR (COMMON)	7	7	7
ASTIGMATISM ADJUSTMENT ELECTRODE	6	20	19
GRID NO. 1	2	14	26

DIM.	LIMITS MIN	MAX
A	25.25	26.0
B	12.38	12.50
C	---	3.75
D	5.5	---
E	17.38	18.13
F	18.5	19.0
G	3.25	3.75

Figure 1 - Outline dimensions of tube type 12ATP28 (cont'd).

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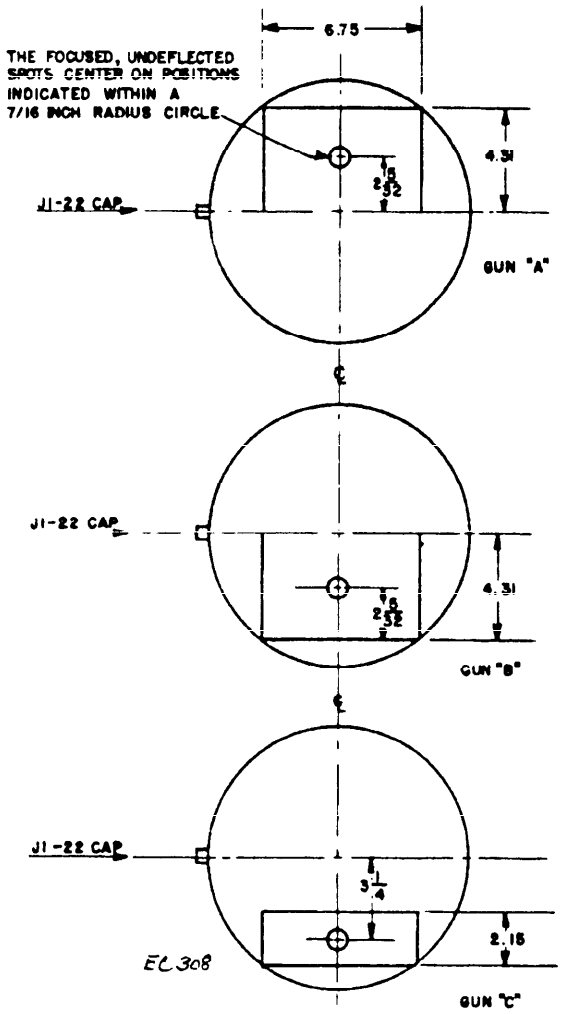


Figure 2 - Useful area of guns A, B and C.

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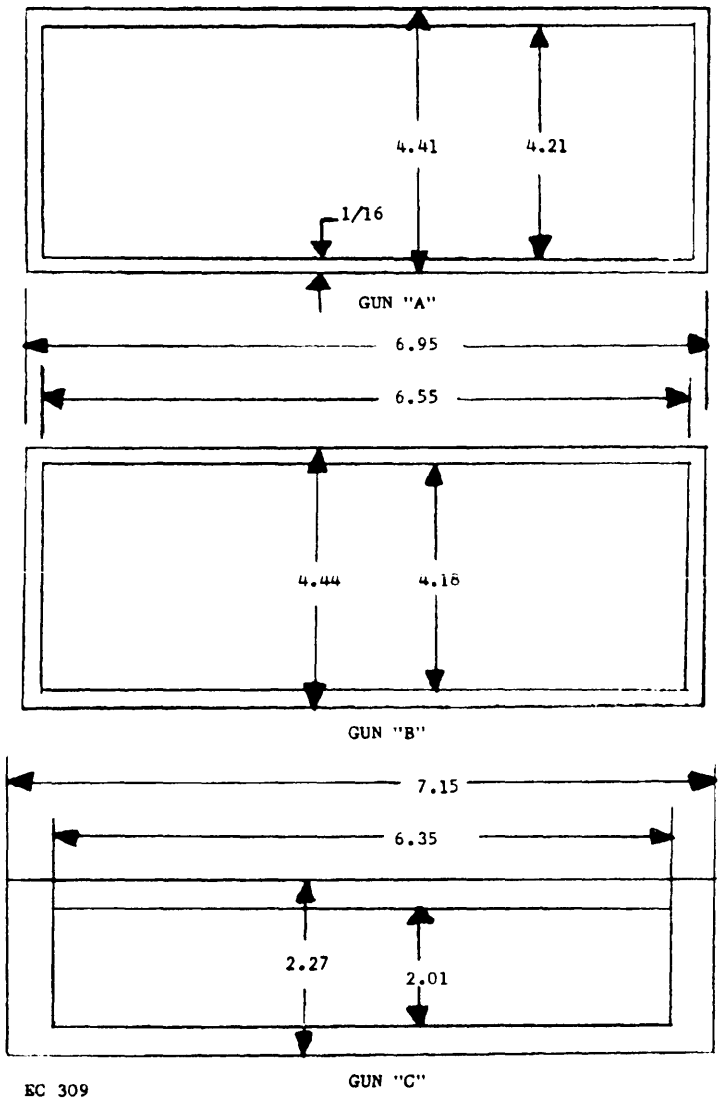


Figure 3 - Pattern distortion diagram.

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NAVAL ELECTRONIC SYSTEMS COMMAND
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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